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324412 (24)

BE (4th Semester)

Examination, April-May, 2013

Branch : Electrical, EEE

ELECTRO MAGNETIC FIELDS

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : Attempt part (a) and any two parts from (b), (c) & (d) of each question. Part (a) is of two marks and parts (b), (c) and (d) seven marks each.

Q. 1. (a) State Maxwell's first equation.

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(b) Transform the vector field $\bar{F} = 2\cos\theta\,\bar{a}_r + \sin\theta\,\bar{a}_\theta$ of spherical coordinates into cartesian components.

(c) Three uniform infinite sheets of charges are located in free space as follows :

(i) 3 nc/m^2 at z = – 4

(ii) 6 nc/m^2 at z = 1

(iii) –8 nc/m^2 at z = 4

Find $\bar{E}$ at points :

(i) P_A (2, 5, –5)

(ii) P_B (4, 2, –3)

(iii) P_C = (–1, –5, 2) and

(iv) P_D (–2, 4, 5)

(d) State and prove divergence theorem.

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Q. 2. (a) What is Electric Dipole ?

(b) Prove that the value of electric flux density at conductor surface is equal to surface charge density. At a point on a conductor surface $\bar{E} = (70\bar{a}_x - 35\bar{a}_y - 70\bar{a}_z)$ V/m, what is surface charge density at that point.

(c) (i) Derive continuity equation of current.

(ii) Derive Laplace's equation.

(d) The potential field at any point in a space containing a dielectric material of $\epsilon_r = 2.1$ is given by

$V = 5x^2y + 3yz^2 + 6xz$ Volts. Calculate volume charge density at point P (2, 5, 3)

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Q. 3. (a) State Biot Savart Law.

(b) State and prove point form of Ampere's Circuit Law.

(c) Magnetic vector potential $\bar{A} = (3y - z)\bar{a}_x + 2xz\,\bar{a}_y$ find $\bar{B}$, $\bar{H}$ and $\bar{J}$.

(d) Express value of $\bar{H}$ in rectangular components at P (0, 0.2, 0) in field of three current sheets :

(i) $2.7\bar{a}_x$ at y = 0.1

(ii) $-1.4\bar{a}_x$ at y = 0.15 and

(iii) $-1.3\bar{a}_x$ at y = 0.25

Derive the formula used.

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Q. 4. (a) Write Lorentz force equation.

(b) The field $\bar{B} = -2\bar{a}_x + 3\bar{a}_y + 4\bar{a}_z$ mT is present in free space. Find vector force exerted on straight wire carrying current of 12A in $\bar{a}_{AB}$ direction where

A (1, 1, 1) and B (3, 5, 6)

(c) Prove that the torque T in closed magnetic circuit can be written as $\bar{T} = \bar{m} \times \bar{B}$.

(d) Explain magnetization $\bar{M}$ and bound current density.

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The magnetization in a certain material for which magnetic susceptibility = 8 is given in a certain region as $150z^2\ \bar{a}_x$ A/m. At z = 4 cm, find magnitude of J_T, J and J_B J_b.

Q. 5. (a) What is Poynting Vector ?

(b) (i) Show that $J_D = \frac{\partial D}{\partial t}$.

(ii) Derive continuity equation from Maxwell's equation.

(c) A certain material has $\sigma = 0$, $\mu_R = 1.$ E = 800 sin $(10^6 t + 0.01z)\ \bar{a}_y$ V/m, find ε and μ using Maxwell's equations.

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(d) (i) Show that the ratio of conduction current density and the displacement current density is $\frac{\sigma}{\omega \epsilon}$ for the applied field $E = E_m \cos \omega t$. Assume $\mu_r = 1$.

(ii) Show that $\bar{E} = -\nabla V - \frac{\partial A}{\partial t}$ where notations have their usual meanings.

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